CHART 6 YDLIDAR X2 LASER OPTICAL PARAMETERS

Item	Min	Typical	Max	Unit	Remarks
Laser wavelength	775	793	800	nm	Infrared band
FDA		⚠ Class I			

Fig. 3: FDA rating of YDLIDAR X2 (Image credit: Lidar datasheet document)

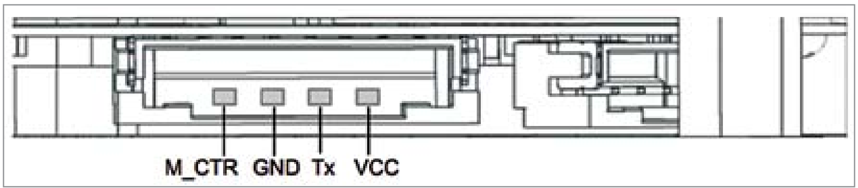

Fig. 4: Lidar pinouts (Image credit: Lidar datasheet)

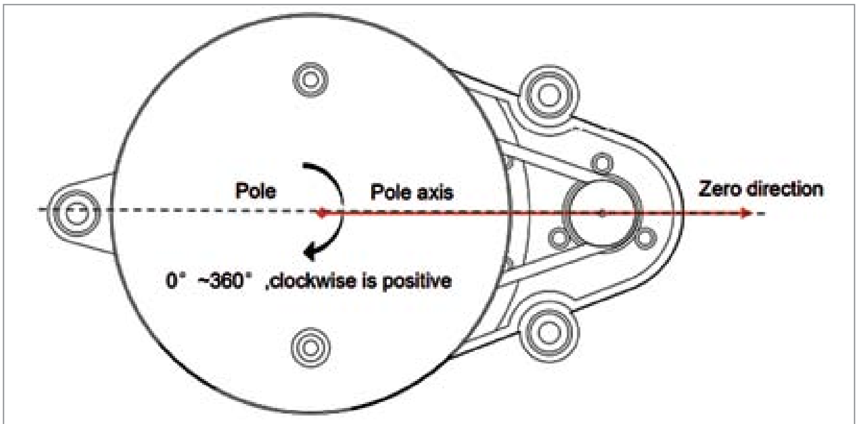

Fig. 5: Lidar's Zero direction (Image credit: Lidar datasheet)

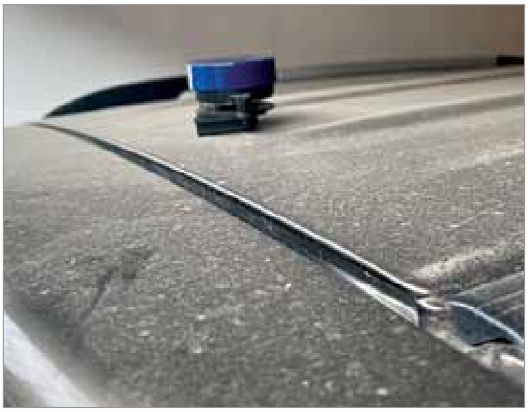

Fig. 6: Mounting of lidar on car's top

solution that works even in a foggy or rainy environment, we need to select the lidar accordingly.

In our design we have used laser based YDLIDAR X2, whose parameters are given in Table 2 and whose image is shown in Fig. 2.

The third thing you need to look into while selecting the lidar is the government's policy and any health risks that may be associated with it. The technology used for the lidar might cause a health risk for the animals, children, or even the plants. The approved FDA parameters of YDLIDAR X2 are shown in Fig. 3.

Circuit diagram

There is no specifice circuit diagram as such for the system. Depending on the model of lidar you are using, the connections might vary slightly. Most 360-degree laser based lidars use serial communication and provide a USB port to connect to the system.